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According to Micron application note TN-00-06. Solve the amount of decoupling capacitors required for a ficticious IC DRAM memory:

VDD = 3.3V + 0.3V/-0.165V

Temp. Range: 0 – 50°C

Currents for each output and n simultaneously outputs:

$C := 25\ pF$ *Load capacitance

$dV := 3.3\ V$ *Absolute voltage range to drive C

$dt := 1\ ns$ *Time to reach dV

$$I := \frac{C \cdot dV}{dt} = 0.0825\ A$$

$n := 64$ *Number of outpus firing simultaneously

$$I_N := n \cdot \frac{C \cdot dV}{dt} = 5.28\ A$$

Maximum reactance at the supply pins:

$\Delta V := 0.05\ V$ *Total change in voltage across the supply pins

$$X_{max} := \frac{\Delta V}{I_N} = 9.4697\ mohm$$

Knee frequency:

$T_r := 1\ ns$ *Rise time

$$F_{KNEE} := \frac{0.5}{T_r} = 500\ MHz$$

Total induntance circuit can tolerate:

$$L_{TOT} := \frac{X_{max}}{2 \cdot \pi \cdot F_{KNEE}} = 0.003\ nH$$

Number of capacitors N needed in the array

$L_{SERIES} := 1.5\ nH$ *Typical series inductance

$$N := \frac{L_{SERIES}}{L_{TOT}} = 498$$

Capacitors value

$L_{SERIES} := 5\ nH$ *Typical series inductance of electrolytic for board-level bypass

$$F_{BYPASS} := \frac{X_{max}}{2 \cdot \pi \cdot L_{SERIES}} = 301\ kHz$$

$$C_{ARRAY} := \frac{1}{2 \cdot \pi \cdot F_{BYPASS} \cdot X_{max}} = 55.7568\ \mu F$$

$$C_{ELEMENT} := \frac{C_{ARRAY}}{N} = 112.0451\ nF$$

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Part number selection

The selected capacitor is the TDK CGA2B1X7R1A224K050BC.
https://product.tdk.com/system/files/dam/doc/product/capacitor/ceramic/mlcc/catalog/mlcc_automotive_general_en.pdf
https://product.tdk.com/en/search/capacitor/ceramic/mlcc/info?part_no=CGA2B1X7R1A224K050BC&utm_source=chatgpt.com

Main specifications:
Nominal capacitance: 220 nF
Capacitance tolerance: ±10%
Rated voltage: 10 VDC
Dielectric: X7R
Operating temperature: −55°C to +125°C
Package: EIA 0402 (1005 metric)
AEC-Q200 qualified

Electrical model:
ESL: 0.370 nH
ESR: 0.0125 Ω

Número de parte del fabricante	Cantidad disponible	Precio Precio por cantidad	Serie	Paquete	Estado del producto	Capacitancia	Tolerancia	Voltaje nominal	Coeficiente de temperatura	Temperatura de funcionamiento
 CGA2B1X7R1A224K050BC CAP CER 0.22UF 10V X7R 0402 TDK	302,347 En stock Este producto tiene un límite de compra máximo.	1 : $0.12000 Cinta cortada (CT) 10,000 : $0.01834 Cinta y rollo (TR)	CGA	Cinta y rollo (TR) Cinta cortada (CT) Digi-Reel®	Activo	0.22 µF	±10%	10V	X7R	-55°C ~ 125°C

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